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ORIGINAL ARTICLE

European cybersecurity disclosure practices: a textual analysis

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Abstract
This study analyses cybersecurity disclosure practices of publicly listed companies in the UK, France, and Germany that have reported cybersecurity incidents in their annual reports. Grounded in signalling theory, our analysis examines the readability and tone of cybersecurity-related disclosures compared to the broader report, aiming to uncover companies' communication strategies to manage public perceptions. Our findings reveal a contrast: while cybersecurity sections employ more complex language than the rest of the report, the descriptions of incidents are notably clear and standardized. The striking similarity across disclosures further suggests that companies may be engaging in impression management to minimize scrutiny. The tension between regulatory transparency requirements and disclosure practices calls for closer investigation.

Keywords Cybersecurity disclosure practices · Cybersecurity incidents · Readability · Reporting · Signalling theory · Tone

JEL Classification G38 · M41

Introduction

Cybersecurity[1] disclosure (CSD) has gained increasing scholarly attention from scholars in recent years (Cheong et al. 2019; Jiang et al. 2022). Much research has focused on regulatory frameworks established by authorities like the Securities and Exchange Commission (SEC) and the European Securities and Markets Authority (ESMA) to mitigate information asymmetry and protect stakeholders through enforced transparency. The SEC issued key statements in 2011, 2018, and 2023, standardizing disclosures via Item 1C of Form 8-K and requiring incident reports within four business days. Similarly, since 2019 European regulatory initiatives have evolved significantly. ESMA (2019) has been issuing guidelines under the Prospectus Regulation to encourage European companies to disclose specific risk factors, including CS and non-financial information (Directive 2014/95/EU (referred to as the "Non-financial Reporting Directive"—NFRD); Directive 2022/2464/EU (referred to as the "Corporate Sustainability Reporting Directive"—CSRD)). Since 2023, the implementation of the updated Network & Information Security Directive (NIS2) and the Digital Operational Resilience Act (DORA) broadens CS obligations to more sectors, enhances security standards for more resilience, and mandates stricter incident reporting. The General Data Protection Regulation (GDPR) has also intensified, leading to increased investigations and sanctions, prompting organizations to improve transparency and embed data protection into their disclosure practices. Additional complementary EU requirements like the Cyber Resilience Act that defines security as a product quality standard or the upcoming Product Liability Directive will hold manufacturers accountable for insecure software. By 2025, cyber risk and digital resilience are set to become regulatory priorities

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[1] Cybersecurity is often considered synonymous with information security (e.g. AICPA (2017); Alexei and Alexei (2022); Haapamäki and Sihvonen (2019)), which focuses on ensuring the continuity of business operations by minimizing security incidents and mitigating the impacts of cyber risks on organizations.

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across the European Union. Nevertheless, European regulations remain less stringent than the SEC's.

A contrast remains despite regulatory initiatives to enhance transparency: while regulators push for more comprehensive CSDs, companies often seem to withhold information on these issues. Although the number of cybersecurity incidents (CSIs) has sharply increased since 2019, there has been a decline in reporting through filings such as SEC Forms 8-K and 10-K (IT Governance 2024). This trend undermines regulatory efforts to improve transparency. Reporting practices vary widely—22% mention incidents under "Risk Factors", 11% in "Footnotes to Financials", and 9% in "Management Discussion & Analysis"—further hindering transparency (Audit Analytics 2022).

Despite the growing regulatory focus, empirical research on CSD practices remains very limited (Cao et al. 2023; Frank et al. 2023; Gauch and Quick 2025; Navarro and Sutton 2024; Perols 2024). Most studies have concentrated on US firms, leaving a significant gap in understanding how European companies handle these disclosures. To fill this gap, we conduct a comprehensive analysis of CSDs and related incidents in the annual reports of large European companies. Using signalling theory, we more specifically examine the readability and tone of CS-related sections compared to the rest of the report, aiming to uncover deviations in communication practices and their implications. Our study concentrates on the HDAX (Germany), FTSE 100 (UK), and SBF 120 (France) as they represent three of the most significant economies within Europe, providing a comprehensive and diversified perspective of the European market landscape. We assess readability using eight indices,[2] narrowing down to the Flesch Reading Ease Test and Gunning Fog index for robustness. Additionally, we examine the tone of CS-related sections and apply a similarity analysis method to identify industry trends, offering insights into corporate communication practices and their implications.

Our analysis reveals a striking contrast: while the language used in CS sections of corporate annual reports is often complex and challenging to understand, the descriptions of specific CSI are comparatively clear and standardized across companies. This suggests that European firms prioritize transparency when reporting experienced CS breaches, even as their broader CSDs remain opaque. Regarding the latter, our findings indicate that companies frequently employ impression management, obscuring information through reduced readability and complicating efforts to enhance overall transparency regarding CSD.

Our study makes three key contributions. Firstly, it extends the understanding of CSD practices by introducing a European perspective**,** offering a novel contribution to the mostly US-centric literature. By examining European firms, we highlight distinctive regulatory and communicative nuances, significantly enriching the geographic scope of existing research**.** Secondly, our paper highlights the contrast in CSD: while descriptions of CSI are comparatively clear and standardized, broader CS sections remain rather complex and difficult to understand. This finding helps to improve disclosure practices by promoting clearer, more accessible reporting frameworks. It helps stakeholders assess risk, provides guidance to companies on improving transparency, and informs regulators' efforts to strike a balance between completeness and readability. It also opens avenues for future research on the drivers and consequences of disclosure complexity and provides insights into effective risk communication and corporate accountability. Additionally, it opens avenues for future research to explore further the drivers and implications of such disclosure complexity, offering deeper insights into effective risk communication and corporate accountability within European contexts. Lastly, it underscores how companies use impression management, obscuring important CS information through reduced readability, complicating efforts to achieve full transparency regarding CSD.

Our findings suggest that companies should improve the readability of disclosures in CS and adopt more comprehensive and transparent reporting practices, if necessary, through more stringent regulations imposed by authorities. This would also be beneficial to shareholders who, when understanding disclosure norms, will be more competent in assessing companies' cybersecurity management and thereby make better investment decisions. Lastly, the complexity of CS sections needs to be interpreted by experts and should not lead to an underestimation of security risks by external stakeholders. This contradictory trend calls for further examination of the strategic drivers behind corporate CS communication practices.

The remainder of the paper is organized as follows: Sect. "Background" reviews the literature on how companies use annual reports for signalling and explores the readability of these reports alongside existing knowledge on CSD. Section "Sample, data, and methodology" outlines our methodology and collected data for analysing communication, followed by the presentation of results in Sect. "Results". Section "Discussion" discusses our key findings. Section "Contributions, limitations, and future research" addresses contributions, limitations, and directions for future research, while Sect. "Conclusion" concludes the paper.

[2] For a discussion of the numerous textual analysis methods, see Bochkay et al. (2023).



Background

Regulatory framework of CSD

Research on corporate disclosure documents, particularly when conducted from a perspective of the companies preparing them, often utilizes two key approaches—signalling theory and impression management theory—when addressing information asymmetry between managers and stakeholders (Merkl-Davies and Brennan 2007). Signalling theory (Spence 1978) suggests that corporate disclosures help reduce information asymmetries by transmitting credible information to minimize uncertainty. Accordingly, managers are expected to send reliable signals that provide stakeholders with incremental, valuable insights. This signalling is typically voluntary, helping companies strengthen trust and maintain stakeholder relationships (Diong et al. 2018; Six et al. 2010; Vosselman and van der Meer-Kooistra 2009). Conversely, impression management theory (Merkl-Davies and Brennan 2007) posits that executives may use disclosures opportunistically, exploiting information asymmetries through biased reporting. Here, the focus is on how information is framed to maintain a positive corporate image, often reflecting opportunistic strategies rather than genuine transparency. This is particularly relevant when managers aim to obscure unfavourable information, such as poor performance, by reducing the readability or quality of the report (Clatworthy and Jones 2001; Courtis 1995; Dempsey et al. 2012; Li 2008). Both theories, while distinct, focus on the strategic communication of corporate information. Signalling theory emphasizes the credibility and utility of information, while impression management theory examines how it is presented to influence stakeholder perceptions. Being the primary documents required by regulatory authorities, annual reports play a central role in addressing information asymmetries. They often extend beyond mandatory disclosures to include voluntary information, which can serve as positive signals of the company's situation and value. Disclosure regulations complement financial regulations by promoting transparency, ensuring firms credibly convey information to reduce asymmetries and related uncertainties, ultimately enhancing stakeholder trust.

In addition to signalling and impression management theories, institutional theory (DiMaggio and Powell 1983) provides valuable insights into why CSD practices may converge. Institutional isomorphism—driven by coercive, mimetic, and normative pressures—can lead firms to adopt similar linguistic and structural characteristics in their disclosures. Regulatory requirements (e.g. from SEC or ESMA) represent coercive pressures that encourage standardized, factually neutral reporting on CSI. Simultaneously, in response to uncertainty and heightened scrutiny, firms may engage in mimetic behaviour, imitating disclosure styles of perceived industry leaders to signal legitimacy and maintain stakeholder trust. Finally, normative influences such as professionalization in investor relations and shared educational backgrounds further promote homogeneity in corporate disclosure practices.

Previous studies report that the implementation of the NFRD and the CSRD is associated with both quantitative and qualitative improvements in non-financial reporting (NFR) across European firms (Arif et al. 2022; Cicchiello et al. 2023). Cicchiello et al. (2023) compare the impact of European regulation on non-financial disclosure (mandatory in the EU, voluntary in the USA) on the Environmental, Social and Governance (ESG) performance of companies. They show that European regulation has a positive effect, significantly improving the ESG scores of the companies concerned compared to those in the USA. Thus, regulatory requirements promote transparency and engagement among European companies regarding social and environmental responsibility. Aboud et al. (2024), also investigate the European's NFRD. They assume this regulatory reduces ESG decoupling—the gap between companies' actual sustainability performance and their reported ESG information. The findings show that NFRD reporting leads to better alignment between what companies do and what they disclose. Overall, the regulation improves the reliability and transparency of sustainability disclosures across the European Union. Breijer and Orij (2022) found that the NFRD has influenced the comparability of non-financial disclosures among listed companies in Europe from 2012 to 2020, with particular emphasis on the adoption of various NFR frameworks. Regulatory increased the use of investor-oriented NFR frameworks but also led to increased information asymmetry among mandatory adopters—companies that disclose NFR until it became mandator, while voluntary adopters decreased the asymmetry. The study also reveals that the country can have a significant influence on how data protection regulations are applied and what disclosure practices are adopted. Sivan-Sevilla (2024) examines the different enforcement strategies employed by national data protection authorities across Europe after the implementation of the GDPR. The study identifies a variety of approaches—deterrence, coercive, educational and persuasion strategies. The findings reveal that there is no single model; strategies depend on factors such as organizational resources, issue salience, and agency independence. In France and UK, the effectiveness and scope of the authorities' monitoring activities depend on their available resources and their level of expertise in data protection. These capacities influence sharply how proactively and broadly they supervise organizations' compliance with data protection laws. Overall, the research highlights significant national divergence

in how GDPR is enforced, despite the common regulatory framework.

ESMA imposes standards such as the Transparency Directive to ensure that disclosures by listed companies are sincere, complete, consistent, and comprehensible. These principles form the foundation for approving annual report publications, aiming to strengthen stakeholder confidence and reduce information asymmetry between companies and stakeholders. Regarding CS, ESMA also mandates specific disclosures with a focus on readability. In its Guidelines on risk factors, ESMA places increasing regulatory pressure on cyber risk disclosures and has been implemented in the EU and associated markets (Georg-Schaffner et al. 2025). Viewing these requirements through the lens of signalling theory, this approach allows companies to send credible signals about their value and resilience to the market. Since 2023, the impact of the NIS2 Directive and DORA has further strengthened CS governance by extending obligations to additional sectors notably critical infrastructure. They further take an increasing focus on operational resilience and requiring a more rigorous incident reporting. Complementary, the Cyber Resilience Act defines security as a product quality standard, while the upcoming Product Liability Directive will hold manufacturers accountable for insecure software. Regarding confidentiality, the GDPR has enforced data residency and protection leading to more investigations and sanctions. All requirements drive organizations to enhance transparency and incorporate data protection practices into their disclosures. These evolving regulatory patterns significantly influence how organizations within the EU develop their CS and data protection strategies, shaping an environment where compliance and transparency are increasingly prioritized in corporate reporting.

Signalling theory has already been employed in CS research to demonstrate the influence of board-level technology committees on firms' ability to detect and respond to security breaches (Higgs et al. 2016) and to analyse board of directors' attributes in understanding CSD (Héroux and Fortin 2024). However, to our knowledge, signalling theory has not yet been applied to the analysis of corporate communications regarding CS and related incidents.

The effects of CSD

Researchers have highlighted the impact of CSD on market indicators such as stock prices, trading volumes, and investor behaviour (Chen et al. 2023), with several studies explicitly examining market reactions to textual risk disclosures in annual reports. Li (2006), when analysing risk words in 10-K filings, found that increased risk sentiment led to decreased future earnings, with stakeholders underreacting to these disclosures. Following new SEC guidelines, Campbell et al. (2014) observed partial investor reactions to cyber risk disclosures, particularly during periods of heightened prominence. Kravet and Muslu (2013) showed that a higher frequency of risk-related sentences in 10-K filings increased investor perception of idiosyncratic risk, affecting stock return volatility and leading to greater forecast dispersion. Collectively, these studies suggest that markets are sensitive to risk information, prompting companies to use textual strategies to disclose this kind of information, often in a "boilerplate" way (Kravet and Muslu 2013). Florackis et al. (2022) conducted a textual analysis of US firms' 10-K CS risk disclosures, identifying a positive trend in the use of quantitative measures. He also observed that industries dependent on information technology reported more quantitative disclosures, which correlated with firms either experiencing or anticipating future cyberattacks.

Following research, markets are highly sensitive to the content of CSD, influencing stock prices, trading volumes, and investor behaviour. These findings suggest that how information is presented plays a key role in determining market reactions. Consequently, the comprehensibility of annual reports—often measured by readability and tone—has been a focus of analysis in the literature. However, while previous studies have examined these factors for various US samples (Calderon and Gao 2021, 2022; Clatworthy and Jones 2001; Courtis 1995; Dempsey et al. 2012; Henry 2008; Li 2008; Loughran and McDonald 2015, 2016), European research on this issue seems surprisingly limited to date. Our paper seeks to fill this gap by analysing whether listed companies in large European countries comply with transparency and comprehensibility requirements set by regulators. Assuming firms report on CS and CSIs in line with these standards, we focus on comprehensibility, reflecting how clearly companies communicate their intent. This criterion is closely linked to the communicative characteristics of the company, requiring the preparer's intent to be clear and understandable. By studying the comprehensibility of CSDs, we aim to provide insights into communication practices, helping regulators to leverage signalling theory more effectively while addressing potential manipulations and misguidance, as suggested by impression management theory.

CSD practices in accordance with comprehensibility requirements

Research on disclosure practices is primarily US-focused and presents a mixed picture. Calderon and Gao (2021) examined how CS risk disclosures in US firms affect auditors' risk assessments and related fees. Using the Fog index—a widely applied readability score based on SEC's plain English guidelines—and disclosure length, they found that lower readability in 10-K reports increased auditor efforts to assess a firm's inherent risk, leading to higher audit fees. Readability characteristics thus suggest that the less



readable and longer a text is, the more time-consuming and expensive it becomes to process the information, driving up costs. Building on their earlier research, Calderon and Gao (2022) further explored the readability of CSDs by comparing the 20-F and 10-K filings of US and foreign firms. Using the same measures—the Fog index and disclosure length—their findings revealed that foreign firms not only disclosed more information about CS risks but also produced more readable reports compared to US firms. In a separate study, Gao et al. (2020) focused on the evolution of CSD readability in public companies, noting that most firms primarily disclosed risks related to service/operation disruptions and data breaches, and found that both the length and complexity of the disclosures increased significantly between 2007 and 2018.

In addition to readability, empirical studies on tone analysis have examined how language is used in CSD and the effects this use has on various outcomes. Using the Loughran and McDonald (2011) dictionaries, Gao et al. (2020) examined the impact of litigious words in reports. They found that such language signals potential legal contests, prompting auditors to increase efforts in assessing the firm's inherent risk, thereby raising audit fees. A litigious tone thus implies higher risk, requiring auditors to integrate CS risk aspects more thoroughly into their assessments. Calderon and Gao (2022) expanded their tone analysis by incorporating uncertainty words alongside litigious text, continuing with the Loughran–McDonald (2011) dictionaries. Their findings showed that foreign firms use fewer litigious and uncertain words to disclose CS risks than US firms.

Other than readability, researchers often analyse narrative tone by measuring the frequency of positive and negative words, a higher ratio of positive terms indicating an optimistic tone (Henry 2008; Loughran and McDonald 2015, 2016; Yan et al. 2024). Disclosure tone, whether optimistic, pessimistic, or neutral, significantly affects investor sentiment and judgement. Research shows that tone offers incremental information by conveying credible insights, helping reduce information asymmetry between managers and stakeholders (Bassyouny et al. 2022). However, managers may manipulate their narrative to create a distorted external perception by adopting an overly positive tone, even if such statements do not necessarily reflect good news (Schleicher 2012). Similarly, they may opportunistically emphasize positive news while downplaying negative aspects to obscure poor performance (Cho et al. 2010; Clatworthy and Jones 2003; Li 2008), driven by overconfidence or a desire to exaggerate performance. As a result, using an inappropriate tone when disclosing information can mislead stakeholders and fail to reduce information asymmetry.

Considering research conducted through textual analysis (e.g. readability and tone) on corporate disclosures, regulation that reinforces or completes with more detailed standards can be expected to effectively address information asymmetries and related costs. This is particularly relevant for CSD, as CSIs significantly affect firms' value creation and investor trust. Consequently, analysing how CS information, especially incidents, is presented in annual reports can provide valuable insights into corporate communication practices. Regulators can leverage these insights to adapt rules, enhancing transparency and mitigating the risk of opportunistic reporting.

Building on this understanding, our study addresses the following research question:

How do European listed companies that have experienced CSIs report on CS practices, and what insights do their disclosure practices?

Unlike previous research focused predominantly on the USA, we compare companies' general CS disclosure styles with their specific reporting on incidents. By examining listed firms from three major European economies, our analysis complements US-focused empirical studies and provides deeper insights into Europe's regulatory landscape and disclosure practices.

Sample, data, and methodology

Our sample comprises 320 firms listed on three major European stock indices: the HDAX (Germany), SBF 120 (France), and the FTSE100 (UK). We manually reviewed the annual reports for the fiscal year 2022 to identify disclosures related to CSIs. This involved a systematic search for specific keywords such as "cybersecurity," "information security," and "incident," followed by a close reading of the relevant sections. We included all firms that disclosed a CSI, regardless of whether the incident occurred in 2022 or in previous years. To ensure data quality, all authors participated in the identification process. In the next step, we extracted and compiled the relevant information on each CSI and its corresponding disclosure into text files for subsequent analysis. In total, 26 of the 320 companies disclosed a CSI: ten in Germany, and eight each in the UK and France.

Using this dataset, we analyse the characteristics of CSD within the annual reports of the 26 European companies that reported CSIs. Our methodology draws on established approaches in non-financial disclosure research, with a focus on three key dimensions: readability, tone, and textual similarity.

Readability analysis

We assess readability using the Python package Textstat 0.4.1, which calculates scores reflecting the difficulty of reading a text. In line with previous studies (Bonsall et al. 2017; Calderon and Gao 2022; Li 2008), we focus on two



widely used metrics: the Flesch Reading Ease Score (1) and the Gunning Fog Index (2).

$$\text{Flesch Reading Ease} = 206.835 - 1.015\left(\frac{\text{total words}}{\text{total sentences}}\right) - 84.6\left(\frac{\text{total syllables}}{\text{total words}}\right) \quad (1)$$

$$\text{Gunning Fog Index} = 0.4\left(\left(\frac{\text{total words}}{\text{total sentences}}\right) + 100\left(\frac{\text{complex words}}{\text{total words}}\right)\right) \quad (2)$$

These scores measure sentence length and word complexity to estimate how challenging the CS sections are for a typical reader. Unlike prior US-focused studies that often use text length as a readability proxy (Berkman et al. 2018; Calderon and Gao 2022), we chose not to include this variable, as our analysis focuses on specific sections rather than entire annual reports.

Tone analysis

We conduct tone analysis using Python to integrate the Loughran and McDonald (2011) dictionaries, a widely accepted approach in financial research (Calderon and Gao 2022; Swift et al. 2020). This method classifies words into categories such as positive, negative, litigious, and uncertainty, generating word counts to evaluate the overall tone of disclosures. Previous studies (Berkman et al. 2018; Swift et al. 2020) have shown that these categories provide valuable insights, often revealing attempts to manipulate the narrative. By analysing the frequency of these words in CS sections, we assess whether companies adopt an optimistic, pessimistic, or cautious tone in their CSD.

Similarity analysis

To assess the similarity of CSD across firms, we use a large language model (LLM) offered by OpenAI (2025) for vectorization and cosine similarity for comparison.[3] This approach captures the contextual and semantic meaning of the text, going beyond simple keyword matching or "boilerplate" detection. While texts contain words, phrases, and sentences, it is difficult to compare them easily. Therefore, text embeddings (numerical representation of words) are a method that is used to transfer texts into vector of floating-point numbers (OpenAI 2025). Those vectors can be compared using different methods, e.g. Euclidian distance, cosine similarity. In accounting research, cosine similarity is frequently used to measure textual similarity (e.g. Bochkay et al. 2023; Brown and Tucker 2011; Gauch et al. 2025; Loughran and McDonald 2016). It measures the angle between the two texts: while small angle between the vectors suggests a high degree of similarity, a large angle suggests a low degree of similarity. Similarity scores range from 0 (no similarity) to 1 (absolute similarity).[4]

In our study, we used a Python script and an API to OpenAI to vectorize the documents. The resulting dataset with embeddings was processed to calculate similarity using the cosine similarity function from the scikit-learn package (Pedregsa et al. 2011).

Results

Overview of experiences CSI/analysis of experienced cyber incidents

In 2022, 26 out of 320 companies in our sample (8.1%)—10 German, 8 French, and 8 UK firms—disclosed a CSI in their annual reports. These incidents, which occurred between 2018 and early 2023, impacted a range of sectors and can be grouped into four categories based on their impact and how companies responded.

Operational disruptions

Eight firms across the three countries reported significant operational impacts due to cyberattacks. In Germany, Aurubis, CompuGroup, Hensoldt, and Sixt faced system shutdowns and interruptions, with Aurubis estimating low single-digit million-euro losses, and CompuGroup hiring external experts to face the issue. In France, Ubisoft, Nexity, and Eurofins encountered similar challenges, with Eurofins receiving a €50 million insurance payout. In the UK,

[3] To vectorize the text we used the LLM "text-embedding-ada-002" provided by OpenAI (2025).

[4] One issue with large language models is that the average cosine similarity between random texts typically ranges from 0.6 to 1 (Liang et al. (2021); Rudman and Eickhoff (2024)). To facilitate interpretation and comparability, we normalized the similarity scores to a standardized range between 0 and 1. Specifically, scores were rescaled such that a value of 0.6 corresponds to 0 on the new scale, while a value of 1 remains unchanged and represents the maximum similarity.

Table 1 Length and ratio of CS and CSI disclosures

	CS Disclosures				CSI Disclosures			
	Min	Max	Mean	Ratio CSD/ Total	Min	Max	Mean	Ratio CSD/ Total
France	728	4199	1801	0.79	20	203	118	0.05
Germany	335	4895	1618	1.36	71	1564	572	0.48
UK	101	3588	1860	0.94	58	2457	535	0.27

This table presents the length of the CSD and CSI measured by words for France, Germany, and the USA. *Min* minium, *Max* maximum

InterContinental also experienced IT system interruptions, though no sensitive data leaks occurred.

Ransomware and data theft

Six companies reported incidents involving ransomware and data breaches. In Germany, Continental dealt with data theft threats leading to potential claims, while ProSieben briefly mentioned minor data leaks. In France, ArcelorMittal subsidiaries faced attacks in 2021 and 2022, prompting investigations without data leaks, and Covivo avoiding significant damage after a 2022 attack. In the UK, Shell briefly reported a 2021 data leak, and NatWest Group noted having experienced several distributed denials of service and ransomware attacks in 2022.

Financial and regulatory consequences

Six firms reported financial impacts due to CSIs. In Germany, Deutsche Telekom provided detailed accounts of attacks on its subsidiary, T-Mobile US, including quantified damages and potential penalties, while Nordex and Evotec experienced delays in financial reporting and temporary stock exchange exclusion. In France, Orange briefly mentioned financial losses from IT incidents without specifics. In the UK, Vodafone documented in a detailed way and disclosed €5 million in costs from a log4j breach and subsidiaries cyberattack, while Weir reported GBP 5 million in costs related to a 2021 incident for consulting, investigation, and hardware.

Proactive CS measures

Six companies highlighted proactive actions after prior incidents. In Germany, Software AG reported improved data security after a 2020 breach. In France, Edenred strengthened its resilience following a 2019 attack, while Bureau Veritas formed an IT Crisis Committee after a 2021 incident. In the UK, Rentokil adjusted its IT management system to address vulnerabilities, and Pearson Group implemented advisory proposals following a 2018 breach, settling with the SEC for USD 1 million. Prudential increased security protocols for new business partners and adapted procedures after a supply chain attack.

Our analysis shows both a broad spectrum of impacts and responses, as well as variation in the length of disclosures. While some companies, like Deutsche Telekom and Vodafone, provided detailed accounts, others such as ProSieben, Orange or Shell disclosed very limited information, usually condensing their descriptions rather than elaborating on the financial effects or post-incident measures taken.

Narrative analysis of annual report

Readability analysis

In line with prior research on CSD, we assess the readability of CS sections in annual reports using the Flesch Reading Ease (Flesch 1948) and Gunning Fog indices (Gunning 1952), along with tone metrics such as litigious and uncertainty words, based on Loughran and McDonald (2011). These variables, widely applied in the literature (Li 2008; Miller 2010; Xu et al. 2018), provide insights into managerial disclosure strategies.

Table 1 presents the word counts of CS-related disclosures across the sample. The highest mean values for these disclosures are observed in the UK, followed by France and Germany. However, when considering the disclosure ratio (CS words as a percentage of the number of words of the entire annual report), Germany ranks highest. Similar trends are found for CSI disclosures. The values show a broad range, indicating substantial heterogeneity in disclosure lengths.

Regarding readability, we use two key measures: the Flesch Reading Ease and Gunning Fog indices. These indices assess reading difficulty by analysing the number of syllables per word and the number of words per sentence. The Flesch Reading Ease score ranges from 0 to 100, where higher scores reflect easier readability.[5] The Gunning Fog

[5] The Flesch Reading Ease score is interpreted as follows: a score of 0-10 is considered extremely difficult, 10-30 very difficult, 30-50 dif-

Table 2 Readability scores of annual report, CS and CSI disclosures

	Flesch reading ease score	Gunning fog index
Annual report	32.165	10.588
CS disclosures	24.290	15.364
CSI disclosures	35.701	15.133

This table presents the readability scores of the CSD and CSI measured by Flesch Reading Ease and Gunning Fog Index

index estimates the years of education required to understand a text, with higher scores indicating more complex and challenging content.[6] Table 2 presents the Flesch and Gunning Fog readability scores for both overall annual reports and CS-related disclosures.

Using the Flesch Reading Ease score, both the overall annual reports and CSDs scored below a level of 50, indicating they are difficult to read and require advanced knowledge for comprehension. General CS disclosures have an average Flesch score of 24, falling into the "very difficult" range and making them significantly harder to read than the overall reports. This suggests that these sections, likely containing more technical language, are less accessible to readers. Interestingly, the readability of CSI disclosures is notably higher, surpassing that of the entire reports. This could imply that firms simplify CSI-related disclosures, potentially to avoid revealing too much sensitive information. The observed lower readability of CSD raises concerns about accessibility and compliance with transparency regulations. From a signalling theory perspective, this complexity does not appear to reduce information asymmetry effectively, potentially undermining transparent communication.

Our Gunning Fog Index results further highlight this disparity in readability. While the overall annual reports score 10.6, falling within the "acceptable" range, CSDs have higher Fog scores—15.4 for CS disclosures and 15.1 for CSI disclosures—indicating these sections are more difficult to comprehend and require a higher level of education or specialized knowledge. This contrast in readability raises questions about the effectiveness of current practices in communicating CS risks and incidents, particularly in meeting regulatory standards for transparency. The high Fog scores suggest that CS disclosures might challenge stakeholders' ability to grasp key information to comprehend substantial CS risks and incidents, which could hinder informed decision-making.

Table 3 Polarity of annual report, CS, and CSI disclosure

	Annual report (%)	CS (%)	CSI (%)
France	− 6.26	− 27.26	− 83.80
Germany	− 0.88	− 25.72	− 45.81
UK	− 0.40	− 23.08	− 62.76

This table presents the polarity of the annual report, CS, and CSI disclosure for France, Germany, and the UK

Table 4 Subjectivity of annual report, CS, and CSI disclosure

	Annual report (%)	CS (%)	CSI (%)
France	5.69	11.87	15.94
Germany	5.73	11.68	10.85
UK	6.20	11.75	12.51

This table presents the subjectivity of the annual report, CS, and CSI disclosure for France, Germany, and the UK

Tone analysis

In addition to readability, we analyse the tone of CSDs in annual reports to understand how European firms communicate CS-related information to stakeholders. Using Python, the tone is assessed along two dimensions: polarity and subjectivity. Polarity reflects the emotional slant, with results indicating whether the disclosure tends towards optimism or pessimism, based on the balance between positive and negative language. A negative result indicates a pessimistic tone, while a positive result reflects optimism.[7] Subjectivity, on the other hand, measures the degree to which the content is opinion-based rather than factual, showing how much of the disclosure is influenced by personal interpretations versus objective facts. Higher subjectivity indicates a more opinion-driven tone, while lower subjectivity signals a more neutral and fact-based approach.[8] Tables 3 and 4 illustrate the results of polarity and subjectivity scores for annual reports, CS-related information, and CSI disclosures of our sample. The

Footnote 5 (continued)

ficult, 50-60 fairly difficult, 60-70 standard, 70-80 fairly easy, 80-90 easy, and 90-100 very easy to read.

[6] According to Li (2008), the Gunning Fog index is interpreted based on the following scale: a score of 14-18 is difficult, 12-14 is ideal, 10-12 is acceptable, and 8-10 is considered too simplistic, often referred to as childish.

[7] Polarity = 100*(Positive words—Negative words) / (Positive words + Negative words). Results can range from -100% (completely negative) to 100% (completely positive), with 0 indicating a neutral tone.

[8] Subjectivity = 100*(Positive words + Negative words) / Total words. Results range between 0 and 100%, where higher values indicate more subjective (opinion-based) content and lower values indicate more objective (fact-based) content.

Table 5 Similarity of CS and CSI disclosures

	France ($N=8$)	Germany ($N=10$)	UK ($N=8$)
Panel A: CS disclosure			
Minimum	0.459	0.279	0.508
Maximum	0.660	0.643	0.749
Mean	0.561	0.522	0.659
Panel B: CSI disclosure			
Minimum	0.230	0.335	0.251
Maximum	0.567	0.650	0.560
Mean	0.399	0.466	0.439

This table presents the textual similarity of the CS, and CSI disclosure for France, Germany, and the UK

analysis reveals a predominance of negative and objective tones across the examined CSDs.

Polarity results indicate that CS-related disclosures are more pessimistic than the overall tone of annual reports, with over 80% of the CS-related content categorized as negative. In France, disclosure around CSIs showed an even more negative tone compared to general CS content, while in Germany and the UK, the tone remains consistently negative across both CS and CSI disclosures. Our findings align with the existing literature, suggesting that negative tones reflect the inherent risks and challenges associated with CS, offering stakeholders a realistic portrayal of the CS landscape and potentially reducing information asymmetry.

Subjectivity results show that CS disclosures are generally more subjective than the overall annual reports. A further notable increase in subjectivity is observed in CSI-specific disclosures, indicating that these sections are framed in a more opinion-based manner. This is particularly evident in France, where CSI-related sections score the highest at 15.94%. The use of a more subjective tone in CSI disclosure suggests companies may nuance their documentation of CS incidents, while broader CSDs are written in a more neutral tone. This difference in tone may hinder the reduction of information asymmetries and impact firms' credibility in reporting CS issues transparently.

Similarity analysis

We further conduct a textual similarity of CSDs and CSI disclosures to identify communication patterns across companies.[9] By comparing the language used, we aim to uncover similarities that may reflect common practices,

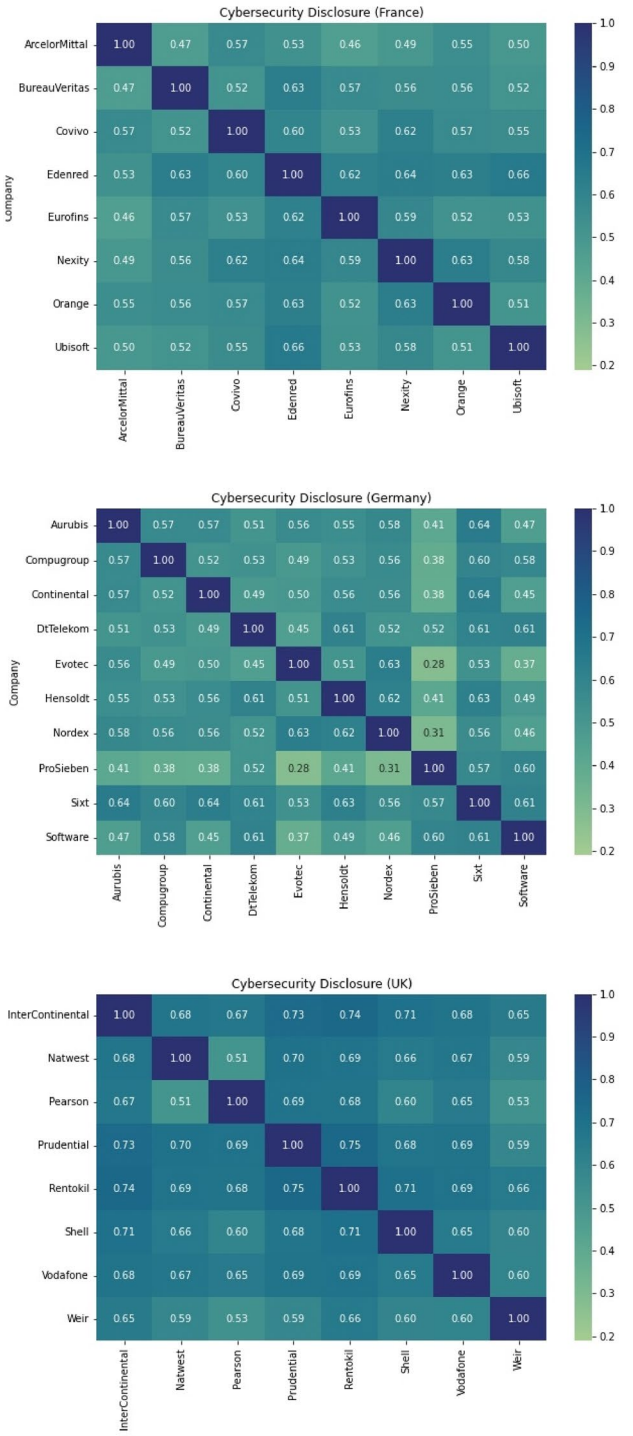


Fig. 1 Similarity of CS disclosures across companies (Model from OpenAI)

regulatory influences, or specific communication strategies. Understanding these similarities helps assess whether CSI disclosures follow standardized patterns or exhibit more individualized, firm-specific approaches regarding CS communication.

[9] Scores range from 0 (no similarity) to 1 (complete similarity), with higher scores indicating more consistent language and structure, while lower scores reflect more diverse, individualized reporting.

Table 5, Panel A presents the results for general CSD across the three countries of our sample. The UK exhibits the highest average similarity, with a mean score of 0.659, suggesting a more homogeneous approach to reporting CS information compared to France and Germany. The visual results of the heatmap (Fig. 1) reinforce this, as the darker shading for UK companies highlights the consistency in their disclosures. French groups also show a relatively homogeneous approach, though to a lesser extent (mean score: 0.561). In contrast, German firms display the most varied disclosures, with a lower mean score of 0.522, and notable outliers like Evoctec, ProSieben, and Software AG displaying less consistent reporting patterns (Fig. 2).

Turning to CSI disclosures, Table 1, Panel B, shows that Germany demonstrates the highest mean similarity (0.466), followed by the UK (0.439) and France (0.399). This variability likely stems from the unique and specific nature of incidents, which leads to more differentiated reporting. Despite the absence of standardized reporting frameworks for CSIs in Europe, it is noteworthy that within-country patterns remain aligned, particularly in the UK, where disclosures appear more homogeneous. This suggests that, even without formal standardization, UK firms adopt more similar approaches when reporting incidents, showing greater consistency in CSI disclosures compared to companies in France and Germany.

Our findings highlight a varying degree of uniformity in CSD and CSI disclosures across the three countries. UK companies exhibit the most consistent reporting, for both general CS and CSI disclosures, likely reflecting stronger adherence to common frameworks or practices. In contrast, German firms show the most heterogeneous disclosures, particularly for CSIs, suggesting less standardized communication practices. French companies display moderate similarity. Overall, the UK stands out for its relatively standardized approach, while both Germany and France demonstrate more diverse disclosure strategies.

In the following, we provide examples of report sections illustrating similarities in reporting between companies.

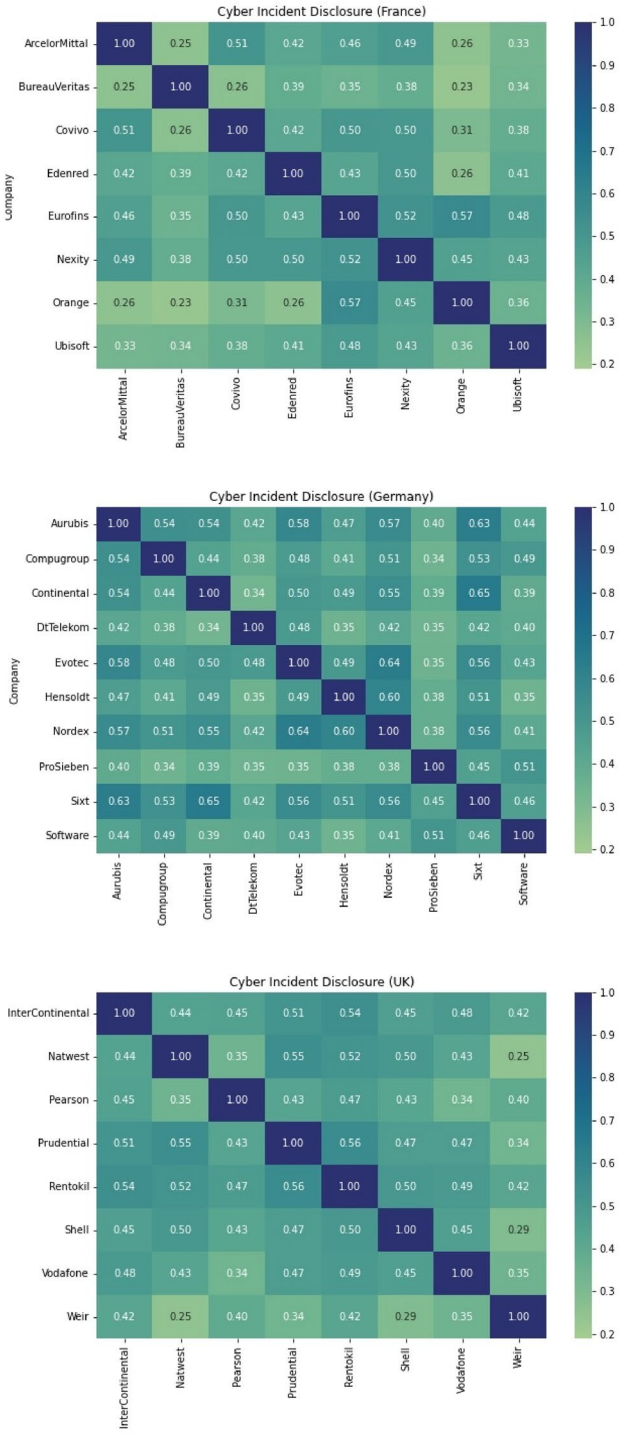


Fig. 2 Similarity of CSI disclosures across companies (Model from OpenAI)

Example 1: The Log4j vulnerability

- "In December 2021, a new critical vulnerability in widely used log4j software code was identified. This vulnerability could be used to steal data, introduce malware or take over systems. The log4j software is used as a building block within many applications and services, and as a result almost all companies were impacted." (Vodafone 2023, p.50)
- "In 2022, the LOG4J or Log4Shell flaw is considered one of the most critical ever discovered. This is a software vulnerability identified and exploited by malicious organisations. Based on data from Tenable, a specialist in vulnerability analysis, as of 1 October 2022, 72% of companies worldwide still had assets vulnerable to Log4Shell." (Nexity 2023, p.60)

Both companies address the Log4j vulnerability, specifically highlighting the Log4j/Log4Shell issue with comparable emphasis and within a similar timeframe. The descriptions of the incident are content-wise similar, focusing on the technical aspects and scope of the vulnerability. Both companies then move to generalize the impact of the vulnerability to the overall economy which is followed by details on the concrete impact to their business. The similarity score between both sentences is 78.750%.

Example 2: Cyberattack with effects on production

- "On 6 April 2023, the Group suffered from a criminal cyber-attack that targeted many of Evotec's operations which caused disruptions to many of its IT systems in several countries and temporarily stopped or reduced the research and production activities." (Evotec 2023, p.65)
- "The Nordex Group was also affected by a cyber security incident at the end of March 2022, the effects of which included production interruptions and in some cases severely delayed installations in Germany during the second and third quarters." (Nordex 2023, p.16)

Both companies share details about a cyber-attack that significantly impacted their production. The information is written in simple language, making it easy for everyone to understand. In the reports, the company furthermore discloses further actions the companies have taken. The similarity score is 67.735%.

Example 3: Cyber insurance

- "Covivio has also taken out an insurance policy against cyber-risks that supplements its insurance cover against the risks of fraud and cyber-attacks." (Covivio 2023, p.113)
- "Additionally, the Group is covered by cyber insurance to cater for a catastrophic cyber event as a risk management mechanism." (Prudential 2023, p.119)

In both reports, cyber insurance is mentioned as part of their risk management strategy. However, there is limited transparency regarding the insurance policies, as key details such as coverage amounts, premiums, and specific terms are missing. Although the companies provide some information, overall transparency could be improved. The similarity score of both statements is 65.50%.

Discussion

Annual reports play a critical role in reducing information asymmetry, and regulators expect firms to clearly communicate their CS posture in these documents. Readability and tone are key indicators of how effectively companies address CS risks and incidents. Deviations in these factors may suggest attempts to impress stakeholders rather than provide accurate, transparent information (Loughran and McDonald 2011). In the USA, despite strict regulatory expectations, CS sections remain complex (Calderon and Gao 2022), while Europe presents a less regulated landscape, leaving room for our analysis of this underexplored area.

Our findings reveal that European firms tend to use highly complex language in CS sections, as reflected by the Fog Index rising from 10.6 (an "acceptable" level) for general reports to 15.4 ("difficult") for CSDs. Moreover, CS sections exhibit Flesch Reading Ease scores as low as 24, falling into the "very difficult" range. This suggests that readers need significant effort, likely requiring higher education or specialized knowledge, to comprehend the content. Such complexity in language can hinder transparency and increase information asymmetry.

Impression management theory (Merkl-Davies and Brennan 2007) posits that managers may strategically frame general and CS-related disclosures differently, with CS content often being framed more negatively due to its inherently risk-based nature. Our tone analysis supports this theory, showing that CS and CSI sections are communicated more negatively than the broader content of annual reports.

Our analysis of CSI reports also highlighted a shift in readability. While general CS sections were classified as "very difficult" on the Flesch Reading Ease scale, CSI disclosures scored in the "difficult" range, using shorter sentences and simpler language. However, the Fog Index for CSI sections remained high, indicating that understanding these incidents still demands a higher level of education or expertise. This contrast suggests that while companies aim to report CSIs more factually, the language complexity may still pose challenges to transparency, limiting the accessibility of information to a broader audience.

A further examination of textual similarity across the reports shows a high degree of uniformity in the wording and style of CS sections, including those detailing specific incidents. This pattern suggests that firms tend to align their communication strategies with peers, possibly to avoid standing out or deviating from perceived norms and expectations. Interestingly, we observe this clustering of language use more within countries, pointing to the influence of cultural and national regulatory environments rather than sector-specific standards in shaping these disclosures.

To further interpret the observed similarities in CSDs, we integrate institutional theory (DiMaggio and Powell 1983), proposing that disclosure convergence results from institutional isomorphism driven by coercive, mimetic, and normative pressures. Regulatory standards from institutions such as the SEC and ESMA represent coercive pressures prompting firms towards standardized disclosures. Additionally, firms facing uncertainty and heightened scrutiny adopt mimetic behaviours, imitating the language and structure of industry leaders to gain legitimacy and stakeholder trust. Normative influences from increasingly professionalized investor relations and CS communication roles further contribute to disclosure uniformity. By introducing institutional theory, we provide a richer theoretical lens that explains the contrast identified in our study, the coexistence of highly complex general CSDs with comparatively clear and standardized disclosures of specific CSIs, beyond impression management alone.

Contributions, limitations, and future research

Our study contributes to the literature by geographically broadening the scope of empirical analysis, explicitly incorporating the three major European economies: Germany, the UK, and France. This geographic expansion is particularly valuable given the dominance of US-centric studies, largely facilitated by standardized disclosure practices associated with SEC filings such as 8-K and 10-K. In contrast, analysing European firms requires a more tailored approach due to varying regulatory contexts and disclosure standards. Identifying companies affected by CSIs and gathering relevant data to construct a comprehensive overview pose greater methodological challenges. Despite these complexities, this geographic enlargement provides novel insights into CSD practices, highlighting distinct regulatory and communicative nuances within the increasingly regulated European environment. By comparing European and US disclosure practices, our findings complement those of Chen et al. (2023), indicating that the level of standardization in disclosure varies significantly across regulatory environments. Despite the relative flexibility in European regulations compared to the USA, we observed a notable degree of textual similarity within countries.

Furthermore, our study applies signalling theory to annual report disclosures on non-financial topics, specifically CS. We posit that companies' disclosure practices serve as signals to the market, shaping perceptions and responses that reflect a company's actual CS stance. Notably, while firms avoid presenting an overly positive view of their CS posture, our findings indicate the use of impression management to obfuscate CS information through reduced readability, thereby making it harder for stakeholders to fully grasp the details. Our research thus also adds to impression management theory, demonstrating how companies divert communication by complicating the readability of negative information to maintain a positive impression on stakeholders. By incorporating institutional theory (DiMaggio and Powell 1983), our study expands theoretical explanations for the convergence observed in firms' CSD practices, complementing existing perspectives such as signalling theory and impression management.

We introduce an innovative methodology by incorporating textual similarity analysis to compare CSD practices across firms and countries. This approach provides valuable insights into common norms and communication strategies. From a managerial perspective, our results suggest that executives and boards should work to improve the readability and objectivity of their CSDs, ensuring alignment with best practices and regulatory expectations. This could also involve drawing lessons from US regulatory environments and adopting strategies that enhance transparency and objectivity, especially when reporting CSIs. Concrete information on revenue loss, penalties, or impact on internal control systems can provide measurable indicators for investors to assess an organization's ability to overcome a CSI. Furthermore, transparency regarding investments in security awareness, employee wages, or cyber insurance coverage can offer insights into an organization's preparedness to face CS challenges. As is the case with DORA, these categories of CS information are of particular interest to critical infrastructures that require protection. However, recent threat analyses show that CSD should concern also small- and medium-sized enterprises and often manufacturers who have been particular impacted by attacks in 2024 (Data Breach Investigations Report 2025). Both sectors and industries have been less regulated in the past but are often core components of the economy and their failure should not be underestimated for customers, the economy, as well as society. As our findings suggest that the situation for large organizations is rather homogenous, we suggest that regulation should also be extended to smaller firm segments.

Finally, for investors our findings provide insights into CSD assessment, enabling better decision-making by identifying firms with more transparent and comprehensive CS reporting. Current assessments conducted by external rating agencies, such as Standard & Poor's or BitSight, rely on non-standardized self-reported data and lack the methodological rigour typically associated with audited annual reports. As a result, investors are often left with limited and potentially unreliable information upon which to base their investment decisions. Moreover, it may be in the strategic interest of organizations to attract investors who make informed, risk-aware decisions aligned with the organization's specific CS posture and risk profile.



One limitation of this study is the relatively small number of firms that disclosed CSIs: only 26 out of 320 companies (~8%) in our sample. This low disclosure rate restricts the extent to which our findings can be generalized across the broader population of European firms. As a result, our analysis focuses primarily on qualitative insights into disclosure practices rather than drawing broad quantitative inferences. Furthermore, the study examines disclosures from a single fiscal year (2022), which may not capture evolving trends or shifts in reporting behaviour over time. The temporal scope thus limits our ability to assess how disclosure practices develop in response to regulatory changes or repeated incident experiences. Finally, although our sample includes companies from three major European indices, the findings may not fully reflect practices in other regions of Europe with differing institutional, cultural, and regulatory contexts. These structural differences could lead to substantially different approaches to CSD than those observed in our study.

Further research should expand the sample size to capture a broader range of European companies across multiple fiscal years. This would help identifying trends and shifts in CSD practices. Moreover, as our findings show that companies tend to use common formats for reporting CSIs, future studies could examine the regulatory drivers behind this standardization—such as the NIS2 or GDPR—and how these regulations influence the content reported to stakeholders. Additionally, research should explore the development of standardized guidelines for CSDs, which could improve readability and reduce information asymmetry. Establishing key elements for CS-related reporting could encourage more uniformity and enhance stakeholders' understanding, while ensuring that companies provide sufficient, clear, and easily accessible information. Furthermore, while much of the prior research has focused on US firms and sections like Items 1A, 1, and 7 in 10-K reports where CS risks are disclosed (Calderon and Gao 2022), recent SEC requirements introduced a new section for CSD titled "Item 1C" in 2023. While this new section provides stakeholders with relevant and easily accessible information about CS risk disclosure, it remains largely unexplored in the literature. To date, we are not aware of any studies on this new section of the 10-K report. However, this opens an avenue for future research. Also, cultural factors may contribute to tone and similarity. For example, we noticed that French firms displayed the most negative tone in CSDs, while UK and German companies adopted a more neutral or even slightly positive approach. Moreover, CSDs tend to be more subjective and biased, with objectivity systematically decreasing in these sections. This trend is consistent with Vitolla et al. (2019) study, which demonstrates that more individualistic cultures tend to adopt more optimistic tone in corporate communication. However, this initial finding needs to be further analysed with data on cultural factors, e.g. provided through Hofstede's cultural dimensions.

Conclusion

This study uncovers the complexities of CSD practices among European firms, where comparatively low regulatory demands for transparency seemingly causes the use of difficult-to-read language. Our analysis reveals a contrast: while CSIs are reported with relatively clearer and more standardized language, broader CSDs tend to be obfuscated through complex and difficult-to-read language, possibly to manage stakeholder perceptions. This reflects impression management strategies, as firms may be attempting to signal transparency regarding CS, while obscuring key details through complex language. Our findings highlight the importance of regulatory frameworks in shaping disclosure practices, with notable similarities in disclosures within countries, particularly in the UK This trend suggests that companies might align their communication practices with industry peers, potentially compromising transparency for conformity.

To address these issues, regulators and decision-making instances must prioritize improving the readability and objectivity of CSDs. Simplifying documentation language and standardizing reporting practices will enhance the comprehension of stakeholders, reduce information asymmetry, and foster trust in CS practices.

Future research should explore the drivers behind the observed standardization and difference in disclosure practices, e.g. looking at cultural components or examining the influence of EU regulations such as the NIS2 and GDPR. Additionally, developing standardized elements for CSDs could improve readability and greater consistency across firms, benefiting both stakeholders and regulators in their efforts to assess and manage CS as well as related risks.

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Data availability The data are available from the authors upon written request.

Declarations

Conflict of interest The authors report there are no competing interests to declare.

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